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Detection of NDM-1 and VIM Genes in Carbapenem-Resistant *Klebsiella pneumoniae* Isolates from a Tertiary Health-Care Center in Kathmandu, Nepal

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Keywords

Klebsiella pneumoniae · New Delhi metallo-beta-lactamase-1 · Verona integron-encoded metallo-beta-lactamase · Carbapenem · Modified Hodge test · Multidrug resistance · Antibiotic susceptibility test

Abstract

Background: *Klebsiella pneumoniae* is one of the leading causes of nosocomial infections. Carbapenems are used as the last resort for the treatment of multidrug resistant Gram-negative bacterial infections. In recent years, resistance to these lifesaving drugs has been increasingly reported due to the production of carbapenemase. The main objective of this study was to detect the carbapenem-resistant genes *bla*_{NDM-1} and *bla*_{VIM} in *K. pneumoniae* isolated from different clinical specimens. **Methods:** A total of 585 clinical specimens (urine, pus, sputum, blood, catheter tips, and others) from human subjects attended at Annapurna Neurological Institute and Allied Sciences, Kathmandu were obtained in the period between July 2018 and January 2019. The specimens were isolated and identified for *K. pneumoniae*. All *K. pneumoniae* isolates were processed for antimicrobial susceptibility testing (AST) using the disk diffusion method. The

isolates were further phenotypically confirmed for carbapenemase production by the modified Hodge test (MHT) using imipenem (10 µg) and meropenem (10 µg) discs. Thus, confirmed carbapenemase-producing isolates were further screened for the production of *bla*_{NDM-1} and *bla*_{VIM} using conventional polymerase chain reaction (PCR). **Results:** Among the clinical isolates tested, culture positivity was 38.29% (224/585), and the prevalence of *K. pneumoniae* was 25.89% (58/224). On AST, *K. pneumoniae* exhibited resistance toward carbapenems including ertapenem, meropenem, and imipenem, while it showed the highest susceptibility rate against to tigecycline (93.1%; 54/58). Overall, AST detected 60.34% (35/58) carbapenem-resistant isolates, while the MHT phenotypically confirmed 51.72% (30/58) isolates as carbapenemase-producers and 48.28% (28/58) as carbapenemase nonproducers. On subsequent screening for resistant genes among carbapenemase-producers by PCR assay, 80% (24/30) and 3.33% (1/30) isolates were found to be positive for *bla*_{NDM-1} and *bla*_{VIM}, respectively. In the same assay among 28 carbapenem nonproducing isolates, 9 (32.14%) isolates were positive for *bla*_{NDM-1} gene while none of them were

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tested positive for *bla*_{VIM} gene. **Conclusions:** Molecular detection of resistant genes provides greater specificity and sensitivity than those with conventional techniques, thus aiding in accurate identification of antimicrobial resistance and clinical management of the disease.

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Background

Klebsiella pneumoniae is a gram-negative bacterium of Enterobacteriaceae family and is a common cause of nosocomial and community-acquired bacterial infections [1]. *K. pneumoniae*, often termed as a medically important genus, serves as a common opportunistic pathogen causing several types of infections, such as pneumonia, urinary tract infection, and bacteremia [2], and some severe forms of infections such as sepsis, meningitis, and wound infections [3]. It can sustain in hospital environments and environmental surface and can colonize human skin, respiratory tract, and bowels [4].

Antimicrobials are the choice of drug against infections caused by pathogenic microbes. However, therapeutic interventions are often challenged due to the development of multidrug resistance (MDR), led by the persisting exposure of the pathogen to a number of antimicrobials such as extended-spectrum beta-lactams, aminoglycosides, fluoroquinolones, and the most effective antimicrobial carbapenems [5]. The resistant spectra have developed periodically since the 1970s with aminoglycoside resistant *K. pneumoniae* followed by subsequent resistance to third-generation cephalosporins in 1980s and 1990s and resistance to carbapenems in recent years [6]. Carbapenem – once used as a last resort drug for life-threatening infections caused by MDR pathogens and extended-spectrum beta-lactamases producing Enterobacteriaceae – is no longer used due to increasing resistance to this drug and is being replaced by another option, colistin [7–9].

Antibiotic resistance in Enterobacteriaceae, particularly the emergence and spread of carbapenem-resistant Enterobacteriaceae (CRE), has become a global serious threat. Infections caused by CRE are associated with various microorganisms that produce carbapenemase enzymes [10, 11]. Several mechanisms contribute to carbapenem resistance, with carbapenemases being particularly challenging due to their ability to hydrolyze almost all beta-lactams [11, 12]. Different mechanisms are involved in the production of carbapenemases, a class of enzymes capable of hydrolyzing carbapenems and other

beta-lactam antibiotics [13, 14]. Carbapenemases fall into 3 different classes according to their amino acid sequence as follows: Ambler Class A serine carbapenemases (KPC, SME, IMI, NMC, and GES, inhibited by clavulanic acid), Class B metallo-beta-lactamases (MBLs) (IMP, VIM, NDM, SPM, GIM, and SIM, inhibited by metal chelators), and Class D oxacillinase-type carbapenemases (OXA types) [13].

MBL belongs to Ambler class B beta-lactamase which requires a metal ion, usually zinc, for beta-lactam hydrolysis [15] because of the independence on catalysis, which is inhibited in the presence of metal-chelating agents such as EDTA [16]. MBLs are encoded either by genes that are part of the chromosome in some bacterial species (resident MBLs) or by heterologous genes acquired by horizontal gene transfer (acquired MBLs). Acquired MBLs were common in *Pseudomonas aeruginosa* and *Acinetobacter* spp. However, most of them have emerged recently at an increasing rate among the members of Enterobacteriaceae [17]. The first MBL was IMP-1 reported in 1990 in *Pseudomonas aeruginosa* from Japan [18]. Since then, it has been reported in various parts of the world [10, 19, 20]. Among MBLs, the New Delhi MBL is found in Asia with the highest frequency (up to 88%) in NDM-1-producing Enterobacteriaceae [21]. Verona integrated MBL (VIM) enzymes have also been reported in most of the MBLs. A previous study has demonstrated VIM genes as the most frequent gene [22].

Many techniques are used for detecting the production of carbapenemases, from phenotypic to advanced molecular-based techniques [23]. The modified Hodge Test (MHT) has been extensively used as a phenotypic technique to detect carbapenemase activity since it is easily available in clinical microbiology routine settings and the process is recommended by CLSI [24]. Real-time polymerase chain reaction (PCR) is a gold standard method for detection; however, it is expensive and thus not available in laboratories, particularly in resource-poor settings [25].

Although a number of studies are underway to address the burgeoning drug resistance of *K. pneumoniae* worldwide [5, 19, 20], there is a paucity of evidence in Nepal. In Nepal, few previous studies have reported the prevalence of carbapenem resistance in Gram-negative bacteria [26–30]. The prevalence of carbapenem was 7.4% in the Model Hospital, Kathmandu [26]; 8% in Tribhuvan University Teaching Hospital, Kathmandu [29]; 12% in the Human Organ Transplant Center, Bhaktapur [27]; and 27% in Kathmandu University Teaching Hospital, Kathmandu [28]. The increasing predominance of carbapenem-

resistant strains in various settings of Nepal further warrants studies to explore effective diagnostics to aid the accurate isolation and guide the clinical management of *K. pneumoniae* infections. It is therefore critical to build up the existing routine antimicrobial susceptibility test (AST) to screen the resistant bacterial strains in the laboratory. Limited studies of carbapenem resistance have been reported from Nepal with paucity of evidence on CRE dissemination in various diseases and in various settings. The main objective of this study was to determine the prevalence of the carbapenemase enzymes (NDM-1 and VIM) in *K. pneumoniae* isolated from clinical specimens

Materials and Methods

Study Design and Study Sample

This hospital-based cross-sectional study was carried out at the Annapurna Neurological Institute and Allied Sciences, Kathmandu, between July 2018 and January 2019. A total of 585 clinical specimens constituting urine ($n = 252$), sputum ($n = 102$), pus ($n = 69$), blood ($n = 74$), catheter tips ($n = 34$), and samples from other sites ($n = 53$) were processed in this study. These specimens were obtained in a sterile, clean, well-labeled, and leakproof container with no visible signs of contamination and were transported to the laboratory [31]. Sample population in this study was inclusive of individuals from inpatient and outpatient departments of the study hospital. In addition, study subjects were representative of all genders and age-groups. Only those samples from the study subject with written informed consent and adequate demographic information were included in this study.

Culture of Specimens and Identification of the Isolates

All samples were inoculated on blood agar, chocolate agar, and MacConkey agar. In addition to these, urine samples were also inoculated in chromogenic UTI agar. Isolates were identified using microbiological tools and techniques such as morphological appearance of the colonies, staining reaction, and biochemical properties [32, 33].

Antibiotic Susceptibility Testing

The Kirby-Bauer disc diffusion method was utilized to check the sensitivity and resistance of isolates to antibiotics. Isolates were plated on Mueller-Hinton agar (MHA) (HiMedia, India), and their susceptibilities to carbapenems were tested by the disc diffusion method according to the Clinical and Laboratory Standard Institutes guidelines [34]. Inoculums were prepared by suspending a single isolated colony on the nutrient broth and comparing the turbidity with 0.5 McFarland standards after proper incubation. Carpet cultures of the bacterial suspension were performed over the MHA plate. Subsequently, antibiotic discs were placed on the agar using sterile forceps and left for 10 min for proper diffusion. Finally, the plate was incubated for 18 h at 37°C. Organisms were classified as “Resistant,” “Intermediate,” and “Sensitive” based on the diameter of the zone of inhibition as recommended by CLSI [34]. Carbapenem class of antibiotics (ertapenem, meropenem,

and imipenem) was used to check the carbapenemase production. The resistant *K. pneumoniae* isolates in pure culture were inoculated in 1-mL sterile tryptic soya broth and incubated overnight followed by aseptic addition of an equal volume of 20% sterile glycerol. The resulting broth was mixed properly by shaking the well and then stored at –20°C until further processing of the isolates for the MHT and molecular analysis [34].

Screening and Detection of Carbapenemase Production

The MHT was used to confirm carbapenemase enzymes produced in bacterial isolates phenotypically [34]. 0.5 McFarland carbapenem-susceptible indicator strain (*E. coli* ATCC 25922) was prepared in broth or saline and was diluted 1:10 in broth or saline. A lawn of the dilution (1:10) of the indicator strain was prepared on MHA as in routine disk diffusion, and the plate was allowed to dry for 3–10 min. A meropenem disc was placed at the center of the plate, while the test specimen was streaked in a straight line from the edge of the disc to the periphery of the plate. The plate was examined after sufficient incubation at 37°C in ambient air for 16–24 h. A clover leaf-like indentation of the indicator strain growing along the test organism growth streak diffusion zone was noted as a positive test, while the absence of the growth of the indicator strain in the aforementioned pattern was taken as a negative MHT test [35].

Crude Plasmid DNA Extraction

Single isolated colony of *K. pneumoniae* was inoculated in Luria-Bertani broth and incubated overnight at 37°C with aeration using a water bath shaker. The plasmid DNA was extracted using an alkaline hydrolysis method. The extracted plasmids were then suspended in TE buffer, labeled well, and stored at –20°C [36].

Amplification of the Carbapenem-Resistant Genes (*bla*_{NDM-1} and *bla*_{VIM}) by PCR

The carbapenemase genes (*bla*_{NDM-1} and *bla*_{VIM}) were amplified by the PCR test. The primers used were reported previously [37] (Table 1). The reaction mixture for the PCR (5× HOT FIRE-Pol Blend Master Mix Ready to Load, Solis BioDyne, Estonia) was prepared as follows: for the blank, 11 µL PCR water, 13 µL master mix, 0.5 µL forward primer (0.5M), and 0.5 µL reverse primer (0.5M). Preparation for positive control followed 9 µL PCR water, 13 µL master mix, 0.5 µL forward primer, 0.5 µL reverse primer, and 2 µL of positive control. Similarly, for the sample, 8 µL PCR water, 13 µL master mix, 0.5 µL forward primer, 0.5 µL reverse primer, and 3 µL of plasmid DNA were prepared. All these mixtures were added to an individual amplification tubes which made the total volume of 25 µL. The template for PCR amplification in clinical isolates was a plasmid DNA preparation. The PCR entailed processing of samples for 3 min at 94°C for initial denaturation; 94°C for 45 s denaturation; 52°C for 40 s annealing; 72°C for 50 s extension; and final extension at 72°C for 3 min for *bla*_{NDM-1} gene. Similarly, for the *bla*_{VIM} gene, the process followed as 95°C for 5-min initial denaturation; 95°C for 30 s denaturation; 53°C for 30 s annealing; 72°C for 30 s extension; and final extension at 72°C for 10-min final extension [37, 38].

Detection of PCR Products by Gel Electrophoresis

The amplified products were characterized by performing gel electrophoresis on 1.5% agarose gel made in w/v tris-acetate-EDTA. 0.5 µL of ethidium bromide was used in the gel as a tracking dye. The gel was allowed to solidify on the plastic cast with comb. After prop-

Table 1. Specific primers used in the study for the amplification of target gene

Target genes	Primers	Sequence (5'–3')	Amplicon size, bp	Reference
NDM-1	NDM-F NDM-R	GGTTTGGCGATCTGGTTTTTC CGGAATGGCTCATCACGATC	475	Nordmann et al, 2012 [6]
VIM	VIM-F VIM-R	GTTTGGTTCGCATATCGCAAC AATGCGCAGCACCAGGATAG	382	

Table 2. Demographic and clinical characteristics of patients attending at the Annapurna Neurological Institute and Allied Sciences

Characters	Samples, N	Culture-positive	
		n	%
Clinical specimens			
Urine	253	104	46.5
Sputum	102	55	24.5
Wound swab/pus	69	31	13.8
Blood	74	19	8.5
Cather tips	34	6	2.6
Other clinical specimens	53	9	4.1
Total	585	224	
Gender			
Male	305	109	48.7
Female	280	115	51.3
Age-groups, years			
0–15	45	11	4.9
16–45	220	100	44.6
46–60	140	48	21.4
>60	170	65	29.1
Type of hospital ward			
ICU	129	75	33.5
Emergency	104	57	25.5
General ward	185	63	28.1
OPD	167	29	12.9
OPD, outpatient department.			

er solidification, 1 µL of 100 bp DNA ladder was loaded; 3 µL positive control and 3 µL negative control/blank were loaded into the well. 3 µL of PCR amplicons were, respectively, loaded on the wells [38, 39].

PCR Controls

The optimization of the PCR was characterized using the positive control of *K. pneumoniae* isolates carrying the *bla*_{NDM-1} gene and *bla*_{VIM} gene. In addition, blank or negative controls were prepared without the application of any DNA.

Quality Control

Strict adherence to the aseptic procedures was taken during each step of the study. Each batch of media and reagents were processed with strict sterility following CLSI guidelines. Purity plates

were used to ensure that the inoculums used for the biochemical test were pure isolates and to explore whether biochemical tests were performed in aseptic conditions. During AST, quality control was ensured using the control strains of *Escherichia coli* ATCC 25922 and *K. pneumoniae* ATCC 700603.

Statistical Analysis

Data were analyzed using SPSS version 24.0. The associations were explored using the χ^2 test at 5% confidence interval among demographic variables such as gender and age of the subjects.

Results

Distribution of the Samples

Among 585 clinical samples, 38.3% (224/585) showed bacterial growth. Bacterial growth isolated from urine sample was the highest (46.5%; 104/224) followed by sputum (24.5%; 55/224) and pus (13.8%; 31/224). The bacterial growth rate was higher among females (51.3%) than males (48.7%). Among the different age-groups, youth population of the age-group 16–45 years contributed the highest growth (44.6%; 100/224), followed by the age-group >60 years (29.1%; 65/224). Similarly, the highest growth was obtained from the samples collected from ICU units (33.5%; 75/224) followed by the emergency department (25.5%; 57/224) (Table 2).

Distribution of Bacterial Isolates

Gram-negative bacteria were predominant (71.9%; 161/224) compared to gram-positive bacteria (28.1%; 63/224). Among 224 bacteria isolates, *E. coli* was the predominant one (29.5%; 66/224), followed by *K. pneumoniae* (25.89%; 58/224) and *Staphylococcus aureus* (20.1%; 45/224) (Fig. 1).

Antimicrobial Susceptibility Profile of *K. pneumoniae*

Out of 224 bacterial isolates, only 58 *K. pneumoniae* isolates underwent AST. *K. pneumoniae* exhibited maximum resistance toward ceftriaxone (89.7%; 52/58) and cotrimoxazole (87.9%; 51/58), followed by amikacin (82.7%; 48/58), and gentamicin (79.3%; 46/58). Resis-

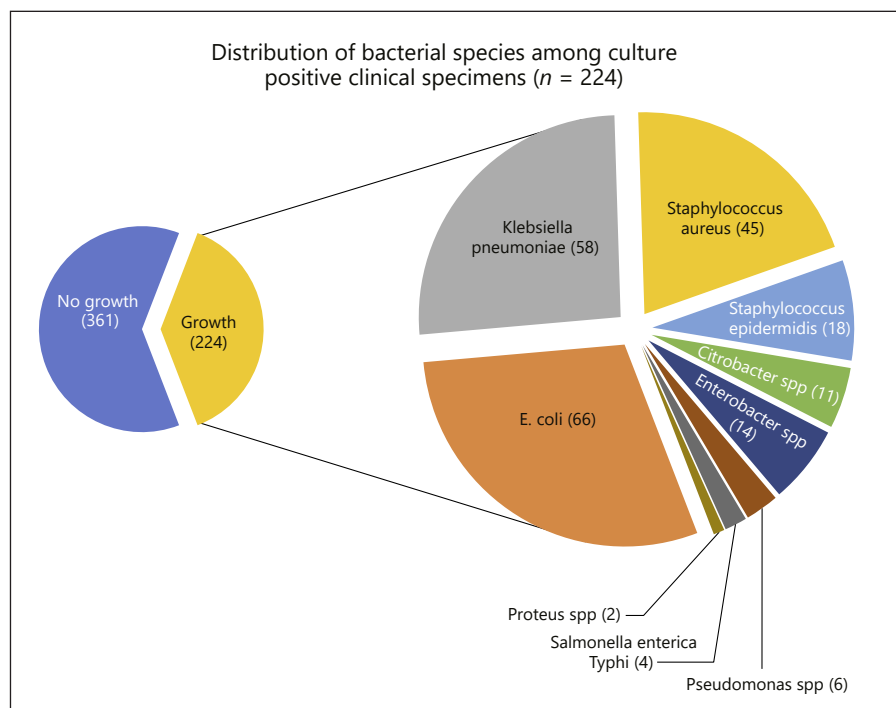


Fig. 1. Distribution of bacterial genera among culture-positive clinical specimens.

tance of *K. pneumoniae* isolates toward meropenem, imipenem, and ertapenem was 60.3% (35/58). *K. pneumoniae* showed the highest sensitivity toward the antibiotic tigecycline (93.1%; 54/58) (Table 3).

Screening for Carbapenem-Resistant *Klebsiella* and Confirmation of Carbapenemase Production

Out of 58 *K. pneumoniae* isolates, 35 (60.3%) showed resistance to meropenem which was screened using a meropenem/ertapenem disc (10 µg). Phenotypic confirmation of carbapenemase production among resistant isolates was done by the MHT using a meropenem disc (10 µg) and *E. Coli* ATCC 25922. In this test, 30 (51.7%; 30/58) isolates were confirmed to be producing carbapenemase.

Pattern of Carbapenemase Production according to Different Clinical Specimens

Out of 30 carbapenemase-positive *K. pneumoniae*, the highest percentage was isolated from sputum samples (76.7%; 23/30), followed by urine (20%; 6/30). There was a significant association between the clinical specimens and carbapenemase-producing *K. pneumoniae* (p value <0.05). The association of carbapenem resistance with different variables such as type of samples obtained, gender, age-group, and indwelling ward/units of the hospital was not significant (Table 4).

Table 3. Antibiotic susceptibility profile of *Klebsiella pneumoniae* (n = 58)

Antibiotics	Sensitive, N (%)	Resistant, N (%)
Amikacin, 30 µg	10 (17.3)	48 (82.7)
Gentamicin, 10 µg	12 (20.7)	46 (79.3)
Piperacillin	11 (18.9)	47 (81.1)
Ceftazidime, 30 µg	15 (25.9)	43 (74.1)
Cefepime, 30 µg	16 (27.6)	42 (72.4)
Ceftriaxone	6 (10.3)	52 (89.7)
Ciprofloxacin, 5 µg	20 (34.5)	38 (65.5)
Cotrimoxazole, 25 µg	7 (12.1)	51 (87.9)
Amoxicillin	20 (34.5)	38 (65.5)
Meropenem, 10 µg	23 (39.7)	35 (60.3)
Imipenem, 10 µg	23 (39.7)	35 (60.3)
Ertapenem, 10 µg	23 (39.7)	35 (60.3)
Tigecycline, 15 µg	54 (93.1)	4 (6.9)

Molecular Screening of Carbapenem-Resistant *Klebsiella* spp. for *bla*_{NDM-1} and *bla*_{VIM} Genes

All phenotypically confirmed carbapenemase-producing isolates underwent molecular screening for detection of resistant genes. Among the 30 carbapenemase-positive isolates, 83.33% (25/30) harbored genes. Of which, 80% (24/30) and 3.33% (1/30) isolates were found positive for *bla*_{NDM-1} and *bla*_{VIM} genes, respectively (Table 4).

Fig. 2. a *bla*_{VIM} gene (382bp) (L, DNA ladder (100–1,000 bp), B, blank; P, positive control; N, negative control; S1–S4, samples). **b** *bla*_{NDM-1} (475 bp) gene (L, DNA ladder (100–1,000 bp); B, blank; P, positive control; N, negative control; S1–S4, samples).

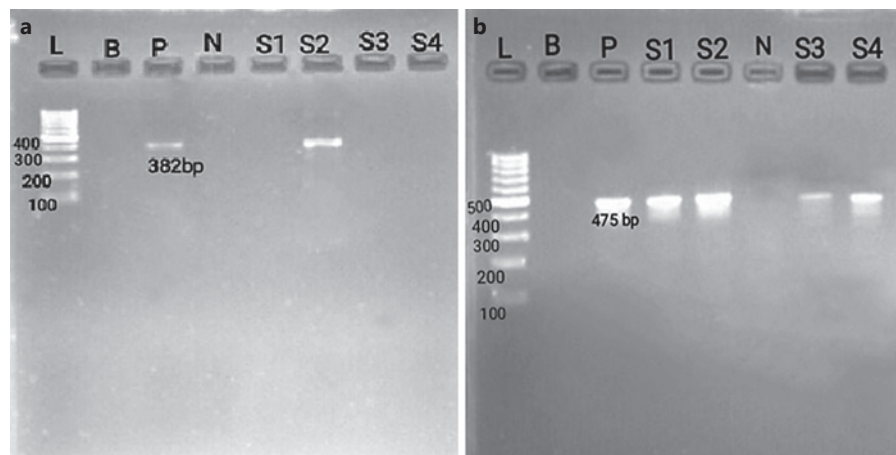


Table 4. Pattern of carbapenemase production by *Klebsiella pneumoniae* according to clinical specimens, gender, age, and types of patients

Characters	<i>K. pneumoniae</i> , N	Carbapenem- resistant** N (%)	MHT (confirmatory carbapenemase test) N (%)	NDM-1 gene N (%)	VIM gene N (%)
Bacterial isolate					
<i>Klebsiella pneumoniae</i>	58	35 (60.3)	30 (51.7)	24 (80)	1 (3.3)
Noncarbapenems producer <i>Klebsiella pneumoniae</i>			28	9 (32.1)	0
Clinical specimens					
Urine	13		6 (20)	3 (12.5)	
Sputum	37		23 (76.7)	20 (83.4)	1
Blood	3		1 (3.3)	1 (4.1)	
Wound swab/pus	4		0		
<i>p</i> value			0.01	0.01	
Gender					
Male	26		12 (40)	10 (41.6)	
Female	32		18 (60)	14 (58.4)	
<i>p</i> value			0.37	0.51	
Age-group in years					
0–15	2		1 (3.3)	1 (4.2)	
16–45	35		20 (66.7)	18 (75)	1
45–60	10		4 (13.4)	2 (8.3)	
>60	11		5 (16.6)	3 (12.5)	
<i>p</i> value			0.15	0.04*	
Type of the patients					
ICU	25		14 (46.7)	13 (54.2)	1
Emergency	7		3 (10)	2 (8.3)	
General ward	9		4 (13.3)	3 (12.5)	
OPD	17		9 (30)	6 (25)	
<i>p</i> value			0.01*	0.02*	

OPD, outpatient department. ** Phenotypically imipenem and meropenem resistant. * Significant.

Molecular Screening of Carbapenemase Nonproducing *Klebsiella* spp. for *bla*_{NDM-1} and *bla*_{VIM}

Among the 28 carbapenemase nonproducing *Klebsiella* isolates, 9 (32.14%) of them tested positive for *bla*_{NDM-1} gene, while none of them were positive for *bla*_{VIM} gene (Fig. 2).

Discussion

Our study was designed to determine the prevalence of the NDM-1 and VIM enzymes that are responsible for the burgeoning resistance toward carbapenems, the last resort drugs against *Klebsiella* species. High prevalence of nosocomial infections in tertiary care centers in low- and middle-income countries such as Nepal are compounded by inappropriate and unregulated use of antimicrobials, leading to the development of antimicrobial resistance (AMR) [40, 41]. AMR is a serious threat to patients undergoing treatment at ICU units with a high prevalence of nosocomial infections. Thus, identification of carbapenemase enzymes responsible for the development of MDR strains among *K. pneumoniae* is critical to inform the clinical management of nosocomial infections [42].

The previous studies have shown the increasing trend of *K. pneumoniae* strains in hospital as well as community settings [43]. In our study, the prevalence of *K. pneumoniae* was 25.8%, similar to some of the studies conducted in Iran [44], Brazil [45], and India [46] and studies from various settings/hospitals of Nepal [42, 47]. In this study, *E. coli* was found to be predominant among bacterial isolates followed by *K. pneumoniae*. This finding was consistent with studies conducted at the International Friendship Hospital, Kathmandu [47]; Padma Hospital, Pokhara [48]; Everest Hospital, Kathmandu [49]; Transplant Center, Kathmandu [27]; Universal College of Medical Sciences, Bhairahawa [50]; Nobel Medical College, Biratnagar [51]; New Delhi, India [52]; Shashemene referral hospital, Ethiopia [53]; and Mexico City, Mexico [54].

Nosocomial pneumonia, a common ICU infection, affects 27% of all critically ill patients, where 86% of it is associated with mechanical ventilation [55]. Our study also showed higher isolation of *Klebsiella* from the ICU unit followed by outpatient department and general ward, respectively. In addition, the highest resistance was found to cotrimoxazole and ceftriaxone followed by gentamycin. Ampicillin and cefotaxime also showed resistance higher than 50%. Although, one of the members of aminoglycosides, gentamicin, showed higher resistance than

amikacin. Comparatively, lower resistance was reported from studies conducted in Guangzhou, China [56]; Tamil Nadu, India [57]; and Gwalior, India [58].

In this study, the overall resistance to third generation cephalosporins and carbapenems was high. These resistance patterns may be due to the acquisition of resistant enzymes such as *bla*_{NDM-1} and *bla*_{VIM}, which can be plasmid-mediated [59]. Our study showed higher resistance to meropenem and imipenem which was comparable to few previous studies from Malaysia [60], Turkey [61], China [56], and India [62]. Effectiveness of an antibiotic can be enhanced when used in combination (for instance carbapenems with colistins) even when the pathogen resists the individual class of antibiotics. Therefore, combination of the drugs can be regarded as an alternative option in treatment regimens [63].

In this study, tigecycline demonstrated high (93.10%) sensitivity against *K. pneumoniae*. Although tigecycline is the last resort antibiotic for carbapenem resistance, the development of resistance is enhanced during the treatment of carbapenem-resistant *K. pneumoniae*; however, its mechanism is not yet understood [56]. One study reported tigecycline nonsusceptibility rate of 2.3% (26/1,116) for *K. pneumoniae* [64]. It is believed that the reduced tigecycline susceptibility of *K. pneumoniae* is the role of RamA on the expression of the multidrug efflux pump AcrAB [65].

Although *bla*_{VIM} was identified since 1997, it represents a minority of carbapenemase in resistant pathogens [66]. In addition, studies focusing on *bla*_{VIM} detection are rare in Nepalese settings. In the current study, the prevalence was 3.33%. The prevalence in our study was comparatively lower than the previous reports from Iran [44, 67, 68], India [46], and Greece [69]. NDM-1 enzymes are capable of hydrolyzing almost all beta-lactam antibiotics (except aztreonam), including broad-spectrum "carbapenems" [23]. Moreover, the genes encoding this enzyme get easily transmitted by plasmid-mediated transfer, leading to uncontrolled growth of resistant strains [70]. Although this study did not determine the co-expression of multiple enzymes, we included NDM-1 strains in addition to VIM. Several studies suggest the prime involvement of these enzymes among resistant isolates in China [71], Greece [72], and Nepal [73–75]. The differences in prevalence may be attributed to the different genetic makeup, spread and outbreak patterns, and acquisition and transfer pattern of resistant genes [68].

Production of carbapenemases is an important mechanism for acquiring resistance by *K. pneumoniae* to carbapenems. Carbapenem-resistant *K. pneumoniae* is resis-

tant to almost all available antimicrobial agents, and infections have caused a high rate of morbidity and mortality especially among the patients from ICU [76]. The majority of these strains are only susceptible to tige-cycline, polymyxin B, and colistin. Beside these drugs, carbapenems are often the drug of choice in most of the health centers.

Real-time PCR is a gold standard method for the detection of the presence of carbapenemase encoding genes. However, it is expensive and therefore not widely available in laboratories in resource-constrained settings. Among various phenotypic methods, the MHT is mostly recommended; nonetheless, several studies have shown poor sensitivity and specificity, particularly for NDM-producers in comparison to VIM producers [77–79]. MHT performance is enhanced by MacConkey agar rather than MHA only in NDM-KP isolates. Performance may be enhanced by using MacConkey agar due to the release of beta-lactamases from the cells in the presence of bile compounds [24]. Our study also showed MHT-positive isolates as PCR-negative and MHT-negative isolates as PCR-positive. This may be the reason that the CLSI 2018 standard excluded the MHT in its guidelines.

Inappropriate and substandard use of antimicrobials can lead to the development of AMR. Development of resistance has also been attributed to bacterial evolution and has been in rise in recent years. For instance, resistance to the first class of antibiotics penicillin G was observed shortly after its discovery. Preventable measures to counteract human contribution to AMR are critical as it can lead to unnatural selective pressure on the bacteria. Multifaceted approaches and collaboration with all stakeholders are essential to curb the current trend of AMR [40].

Clinical and Policy Implications

Infections caused by the resistant strains of gram-negative bacteria are notoriously difficult for management with the available treatment regimens. Moreover, augmented risk of MDR infections basically among female population (who are more prone to common UTIs) and the patients admitted to various wards of hospitals including ICU and general surgery warrant the need of early diagnosis and management of the infectious diseases. Furthermore, finding of our study suggests the need of regular monitoring and surveillance of antibiotic resistance and determining the prevalence of genes harbored by such pathogenic strains across the country. Increasing spectrum of MDR pathogens may cause problems to physicians in choosing the prescription of appropriate thera-

peutic options. Therefore, tertiary care centers should be well equipped with molecular diagnostic facilities in addition to the conventional one, which helps in the proper infection control at the earliest. Additionally, imposing restrictions on over-the-counter use of drugs and implementation of antibiotic stewardship programs in the health-care centers can facilitate and promote the rational use of antibiotics throughout the country.

Strengths and Limitations

This study will be a useful reference for future studies to explore and expand on the wider prevalence of carbapenem-resistant organisms in hospitals, patients, and community settings. There was a paucity of evidence, particularly from Nepal to compare our findings. Nonetheless, this study was based on a single hospital setting with limited study subjects; broader surveillance in many hospitals and communities is essential to establish the precise rate of these isolates and resistant genes. This study detected only *bla*_{NDM-1} and *bla*_{VIM} genes and their possible transferability. Although the MHT was not the best approach in phenotypic determination, this study relied on the same test due to unavailability of newer techniques such as carbapenem inactivation method, carbaNP, and blu carba test. Because of the poor sensitivity and specificity of the MHT, the actual burden of the resistance to carbapenems could be higher than our findings. We recommend future researchers to implement the newer and more reliable aforementioned tests in their studies. In addition, detection of resistant genes and strains of the same and other Ambler classes are recommended for future studies.

Conclusions

Production of carbapenemase by *K. pneumoniae* leads to the development of resistance to the carbapenem group of antimicrobials. The emergence and spread of resistant genes, especially *bla*_{NDM-1} and *bla*_{VIM}, have threatened the treatment options available so far. The presence of MBL gene among carbapenemase nonproducing isolates can act as a reservoir of resistance and can spread within the health-care setting and communities. Molecular detection of resistant genes has more specificity and sensitivity than those of conventional techniques, and thus, adopting these methods for detection can aid in early and accurate identification and clinical management of diseases.

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Statement of Ethics

The Ethical Review Committee of the Nepal Health Research Council approved this research (Regd. No. 545/2018). Written consent was applicable to literate people, while verbal consent was approached for the rest subjects. Parents/guardians were interviewed in case of children. Strict adherence to the ethical guidelines was taken, and we declare that this research is free from selection bias.

Conflict of Interest Statement

All the authors declared that they have no competing interests.

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Author Contributions

All the authors made a substantial contribution to the study. S.T., N.A., I.L., and K.R.R. conceived and designed the study. S.T. collected samples and investigated and recorded the laboratory findings. A.K.S., N.A., I.L., and U.T.S. supervised the laboratory work. B.D., K.R.R., M.R.B., B.A., and P.G. advised and formulated the methodology for the study. B.D. and K.R.R. drafted the original draft. B.D., K.R.R., and B.A. are responsible for reviewing several versions of the manuscript. Others helped to review and amend the manuscript. All the authors read and approved the final manuscript.

Data Availability Statement

All data pertaining to this study are within the manuscript.

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